

Real-time video anomaly detection — 11 classes, open-set

One frozen encoder, encoded once, read three ways. No backbone fine-tuning.

58.8

benchmark score

18–58×

realtime

125 s

all 28 videos

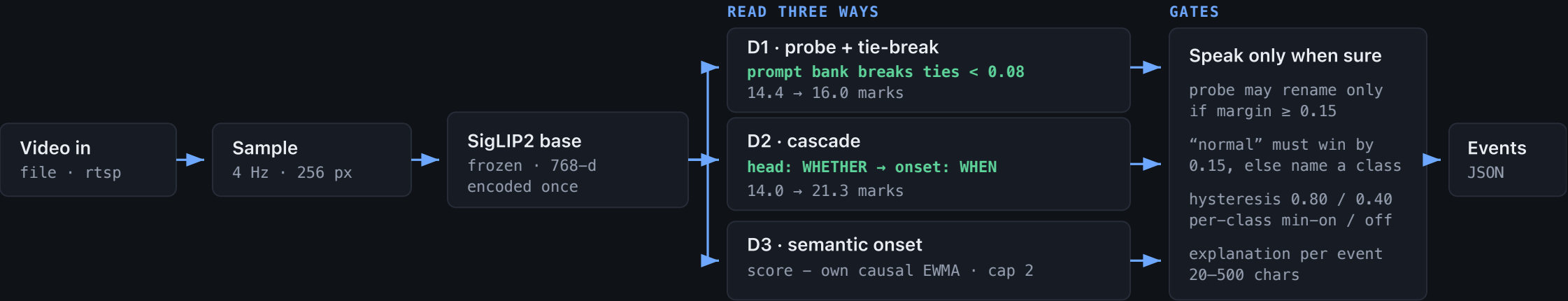
2.4 MB

trained weights

A2000

12 GB GPU

PIPELINE



WHY THIS SHAPE

- **Frozen encoder, one pass.** Features computed once, read three ways — why it beats realtime on a modest GPU.
- **Prompts are data, not weights.** A twelfth class is a sentence, not a retrain.
- **Score against normal.** Margin over the best of ~70 normal prompts; raw similarity just describes the scene.
- **Long video needs a baseline.** D3 keys on the rise over a causal 20 s EWMA, so a busy junction isn't permanently anomalous.
- **Locating ≠ naming.** Detector says when, probe says what — 63% vs 33% on class.

ENCODERS EVALUATED — IDENTICAL SCORING CODE

BACKBONE	DIM	D1	
SigLIP2-base/224 shipped		768	45.8%
SigLIP2-so400m/384		1152	54.2%

Nine full runs, every one scored on the live benchmark

Each row is one complete pipeline run over all 28 videos. Nothing estimated, nothing assembled from parts.

EVERY SUBMISSION, AND WHAT IT SCORED

#	WHAT WE CHANGED	D1 /25	D2 /35	D3 /40	MARKS
1	Baseline — probe · temporal head · onset	11.6	14.0	16.9	42.5
2	"normal" must win by a margin, else name the anomaly	14.4	14.0	16.9	45.3
3	Half-max boundary refinement	14.4	14.0	8.0	36.4
4	Onset instead of the head on D2	14.4	12.6	16.9	43.9
5	Adaptive per-video D2 thresholds	14.4	5.3	16.9	36.6
6	Tighter D2 hysteresis	14.4	14.0	16.9	45.3
7	D2 cascade — head decides whether, onset decides when	14.4	21.3	16.9	52.6
8	Keep 2 intervals per long video, not 3	14.4	21.3	17.9	53.6
9	Prompt bank breaks the probe's ties	16.0	21.3	17.9	55.3

Plus a reasoning bonus of +3.5/5 throughout — **58.8 total**.

WHAT WORKED

- **Run 7, the biggest single gain.** The head matched 0 of 12 events but correctly stayed silent on the one normal video; the onset localised well but fired on it. Chaining them collected both — **14.0 → 21.3**.
- **Runs 2 and 9, both on D1.** One clip read normal 0.132 vs smoke 0.131. Letting a second component break ties like that took D1 **46.4% → 64.1%**.

WHAT DIDN'T

- **Run 3 halved D3.** We tightened intervals to their peak width; the real events run 38–125 s, so tidy spans miss them under tIoU 0.5.
- **Run 5 cost a whole video.** Forcing every clip to fire lost 8.7 marks — almost exactly the 8.75 of the one normal video.

The pattern: every gain came from a routing rule — which component may speak, and when. Every loss came from tuning a number to the data. The three winning changes are about 30 lines of code, no new weights.